

## Reviewer Report

Journal: Conservation Physiology

Manuscript: Xenobiotic Exposure May Confound Traditional Stress Biomarkers:  
Evaluating PFAS and Cadmium Effects on Largemouth Bass Leukocyte Profiles

### Comments to the Author

This manuscript addresses a relevant and timely question for conservation physiology, particularly the potential for xenobiotic contaminants to confound commonly used physiological stress biomarkers. The experimental concept is sound, and the study is generally well written. However, several aspects of the Methods and analysis require clarification to allow full evaluation and reproducibility of the results. In particular, clearer description of leukocyte ratio calculation, exposure preparation and persistence, solvent use, verification of nominal concentrations, salinity verification at the acute sampling timepoint, and statistical model structure would substantially strengthen the manuscript. None of these issues appear insurmountable, and addressing them should improve confidence in the conclusions and enhance the manuscript's contribution to the field.

### MAJOR COMMENTS

Neutrophil-to-lymphocyte ratio definition and calculation:

L221: The calculation of the neutrophil-to-lymphocyte ratio is unclear and potentially inconsistent with standard usage. The Methods state: “We examined smears under a light microscope (40X magnification), identified the first 100 leukocytes, and calculated the NLR by dividing the observed lymphocyte count by the neutrophil count” (Methods, Section 2.3, Neutrophil-to-Lymphocyte Ratio). As written, this describes a lymphocyte-to-neutrophil ratio rather than a neutrophil-to-lymphocyte ratio. Because NLR is the sole physiological response variable used in the study, this point must be explicitly clarified, as it directly affects interpretation of all results.

Experimental unit and replication structure:

L166: The experimental unit and replication structure are not clearly defined. The Methods state: “We assigned each tank to one of five groups... with three replicates per group” (Methods, Section 2.1), which suggests tanks are the experimental units. However, the statistical analysis refers to “a random effect for individual (i.e., tank)” (Methods, Section 2.5). The authors should clarify whether each tank housed a single fish and how repeated measures were handled to avoid ambiguity regarding independence.

Water changes, exposure duration, and contaminant persistence:

L195: The Methods state: “On day 8, we completed partial water changes in all tanks... to maintain water quality” (Methods, Section 2.2), but no additional water changes are

described after contaminant addition on day 10. As written, this suggests continuous exposure from day 10 to day 22. The authors should clarify whether water changes occurred after day 10, whether contaminants were renewed, and discuss expected half-lives of Cd and PFPeA.

Solvent use and compound preparation:

L198: The Methods state: “We applied low and high concentrations of Cd... and PFPeA... directly to the water” (Methods, Section 2.1), but do not indicate whether a solvent was required for compound solubility. The authors should explicitly state whether a solvent was used and whether a solvent control was included.

Verification of nominal contaminant concentrations:

L198-202: The manuscript does not describe how nominal concentrations were verified. Beyond stating that compounds were added “directly to the water” (Methods, Section 2.1), there is no indication whether concentrations were analytically confirmed. This limitation should be clarified, particularly when interpreting null effects.

Salinity stressor verification:

L206: The Methods state: “Three hours after salt introduction, we collected a third blood sample” (Methods, Section 2.2), but salinity measurements are only reported later: “We collected measurements... again on day 17” (Methods, Section 2.4). Salinity at the acute sampling timepoint should be reported or acknowledged as a limitation.

Statistical analysis clarity:

L 248: The authors state: “We used a generalized linear mixed-effects model (GLMM)” (Methods, Section 2.5), but do not specify the distribution, link function, or software. Additionally, the sentence “We assessed model convergence by ensuring the residuals were normally distributed” (Methods, Section 2.5) conflates residual diagnostics with convergence and should be revised.

## MINOR COMMENTS

Interpretation of stress responses:

L409: The Discussion states: “Our results could indicate that fish already experiencing elevated stress may not mount an expected response to an additional stressor” (Discussion, Section 4.4). This interpretation should be framed more cautiously, as alternative explanations such as ceiling effects are possible.

Equating leukocyte ratios with stress:

L 367: The manuscript occasionally equates leukocyte ratios directly with stress, for example: “PFPeA exposure elevated stress in freshwater fishes” (Discussion, Section 4.3). More precise language would improve clarity.

Sample size reporting:

L 239: Mortality is described as “one fish from the control, Cd high, and PFPeA high groups died... one fish from the PFPeA low group also died” (Methods, Section 2.5), but sample sizes per treatment and timepoint are not explicitly reported. A table summarizing sample sizes would improve transparency.

Reporting percent changes:

L292: Percent changes are reported without raw values, for example: “NLR in the PFPeA high group increased by 212%” (Results, Section 3.3.1). Including raw or summary values would improve interpretability.

Abstract wording:

The Abstract states: “PFPeA is a short-chain PFAS chemical commonly used in consumer products,” but it is unclear whether this refers specifically to PFPeA or PFAS compounds more broadly.

Feeding regime description:

L351-352: The Discussion notes: “We restricted food intake during the study” (Discussion, Section 4.2), but feeding frequency and duration are not described in the Methods.